LensJudge: An Agentic Visual-Inspection Grader for Strong-Lens Candidates on the Claude Agent SDK

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Internal Prototype + LensBench-VI Evaluation

Abstract

The Huang group's ResNet/EfficientNet finders emit thousands of above-threshold cutouts per data release; turning them into graded candidates (A/B/C/D) is a human visual-inspection (VI) bottleneck. We build **LensJudge** — the system the group's onboarding report calls "LensAgent"/"VI Pre-grading Agent", renamed to avoid a name clash with an existing arXiv paper — on the *Claude Agent SDK for Python*, and ask whether the data on hand can *test* such a system. The answer is yes for imaging: ∼2,706 human-graded A/B/C cutouts, ∼2,345 Grade-D human-*rejected* hard negatives, ∼65K random-galaxy negatives, and a Foundry-II confirmed/non-lens gold tier are all in hand. We implement three graders — a lean single multimodal call, a perspective-diverse judge panel, and a factor-decomposition multi-agent design — plus a spectroscopic pre-grader for discordant-redshift pairs, and score them in a shared harness (LensBench-VI) on a frozen held-out set of 200 imaging systems. Because every label is a single 2-author consensus (no per-rater data), all agreement numbers are *consensus-referenced with no human ceiling*. Two results stand out. (1) On the *spectroscopic* task — clean physical features, spectroscopically confirmed gold — the agent is strong: 20/20 Hsu Table-2 Grade-A pairs re-graded plausible, 4/4 Foundry-II non-lenses rejected. (2) On *imaging* grading of the CNN's hard high-score pool, no configuration — lean Sonnet, lean Opus, a judge panel, or the §9.1 multi-agent design — reproduces the single-consensus A/B/C grades: all four are near-chance (AUC 0.32–0.51) and uniformly skeptical, and paying 2–4× more per candidate buys nothing. We argue this is less a model failure than a measurement problem — A/B/C are mostly *unconfirmed* candidates and the C/D boundary is one rater's subjective call — which turns the missing multi-grader ceiling from a caveat into the central finding.

1 Motivation

Lens finding in wide-area imaging is a two-stage pipeline: a CNN ranks $O(10^7)$ galaxy cutouts, and humans visually grade the $O(10^3–10^4)$ above-threshold survivors into A (clear lens), B (probable), C (possible), and D (rejected) (Huang et al., 2020, 2021; Inchausti Reyes et al., 2025; Storfer et al., 2024). The VI step is the wall-clock bottleneck and will only worsen at Rubin scale. LensJudge automates the structured judgment of that step with a tool-using LLM agent, pre-grading candidates and escalating the genuinely ambiguous ones to a human — it is a triage assistant, not an autonomous arbiter.

2 Do we have enough data to test it?

Imaging: yes, today. From the group's reproductions we assemble (all public / on-disk): *positives* — 2,706 graded A/B/C candidates as $101 \times 101 \times 3$ *grz* FITS (Storfer DR9 + Inchausti DR10) with per-candidate CNN probabilities; *hard negatives* — 2,345 Grade-D human-rejected candidates (the false positives the CNN flagged but humans rejected) parsed from the published catalogs, cutouts fetched on demand; *easy negatives* — ∼65K random-galaxy cutouts; *gold* — 21 Foundry-II confirmed/known lenses + 4 confirmed non-lenses (Huang et al., 2025). **Spectroscopy: catalog-level today.** Hsu's 13,530 discordant-redshift fiber pairs and 20 Table-2 Grade-A systems (Hsu et al., 2025) give a spectroscopic test set, though the raw DESI fiber flux is not local.

Two honest gaps. (1) Every grade is a *single* 2-senior-author consensus — there are no per-rater labels anywhere — so a human-vs-human Cohen-κ ceiling cannot be computed; we report agreement, not accuracy, and flag a ∼250-cutout multi-grader study as the single highest-value future addition. (2) The spectroscopic grader reasons over catalog features (z_l, z_s, separation, σ_v, θ_{E}) plus DR10 imaging; raw-flux reading (SPARCL / public healpix coadds) is a documented enhancement.

3 Architecture on the Claude Agent SDK

The SDK has no image-in-prompt, so the only way the model sees pixels is a tool that returns an image content block. Our `fetch_cutout` tool renders the *grz* cube to multi-view Lupton-RGB PNGs (full / center-zoom / lens-light-residual / high-contrast, identical params to the group's inspection viewer) and returns them; `get_photometry` measures aperture $g-r, r-z$ colors; `crossmatch_local` flags prior-catalog overlap; `get_specfit` does the SIS θ_{E}/separation check. Tools are an in-process `create_sdk_mcp_server` bundle; because the Python `@tool structuredContent` is not forwarded, the grade contract is a system-prompt JSON schema validated into Pydantic models with one repair retry; PreToolUse/PostToolUse hooks stream a per-candidate JSONL trace (with `total_cost_usd`) that doubles as the eval harness.

We implement three imaging graders of increasing cost: **lean** (one multimodal call; the five Huang-2020 criteria as scored fields of a single integrated judgment), **panel** (four perspective-diverse judges — Advocate, Skeptic, Morphologist, Contaminant-hunter — fused by ordinal quorum + skeptic-veto + a 2-of-N A-rule), and **multiagent** (the onboarding-plan §9.1 factor-decomposition: morphology/color/ contaminant specialists → GradeArbitrator). A **spectro** grader classifies discordant-redshift pairs lens/dimple/not_lens.

4 LensBench-VI methodology

The held-out set is a frozen, hash-pinned manifest of 200 systems: graded A/B/C positives + Grade-D hard negatives + random-galaxy negatives + Foundry-II gold, stratified by grade and region, negatives kept as separate strata. The LLM grader never trained on any cutout, so all rows are valid held-out for it (the NeuraLens training-set leakage that affects the frozen-CNN baseline is recorded per row). Tasks/metrics: (i) binary lens(A/B/C) vs non-lens(D) — ROC-AUC and recovery at 1%/0.1% FPR (the group's operating point); (ii) ordinal A/B/C/D vs consensus — quadratic-weighted κ, exact/adjacent accuracy, confusion; (iii) calibration of p_{lens} (ECE, Brier); (iv) agent-vs-CNN agreement. Baselines: the p_{meta} threshold, single nets, majority-class, and image-only-no-tools.

5 Results

On the full 200-system clean-imaging held-out set the lean Sonnet grader scores binary ROC-AUC 0.51, recovery@1% FPR 0.04, quadratic-weighted $\kappa = -0.02$, ECE 0.42 at \$0.06/candidate — i.e. essentially no agreement with the single-consensus A/B/C labels. Table 1 asks whether a stronger backbone or multi-agent deliberation recovers it, on a shared 48-system slice. All numbers are **consensus-referenced, no human ceiling**.

Table 1: Grader comparison on the shared 48-system imaging slice (consensus-referenced; no human ceiling). No configuration separates from chance.

Metric	lean-Sonnet	lean-Opus	panel	multiagent
Binary ROC-AUC	0.44	0.32	0.42	0.38
Quad-weighted κ	-0.13	-0.06	-0.07	-0.13
Mean cost / candidate (\$)	0.06	0.13	0.25	0.20

Key finding — on the hard imaging pool, neither the agent nor the CNN separates, and more compute does not help. The frozen ensemble cannot separate real lenses from the Grade-D human-rejects (mean $p_{\mathrm{meta}} \approx 0.82$ on rejects, near its value on lenses) — but, crucially, *nor does the agent*: every configuration sits at AUC 0.32–0.51 and $\kappa \approx 0$, assigning $p_{\mathrm{lens}} \approx 0.02$–$0.03$ even to consensus-A systems, and a 2–4× more expensive Opus / panel / multi-agent run does not improve it. The agents are *uniformly skeptical* of the whole high-p_{meta} pool.

Why this is the right answer, not a bug. We verified the cutouts are real and correctly resolved (all on disk; obvious arcs, e.g. `DESI-091.7214−58.9787`, do get high p_{lens}). The pool is the hardest possible slice — positives and CNN-rejected negatives *both* have high p_{meta}— and on it the A/B/C labels are mostly *unconfirmed* candidates whose C-vs-D boundary is one rater's subjective call. So an agent that is skeptical of unconfirmed C-grade candidates is not demonstrably wrong; it simply cannot be *adjudicated* against a single-consensus label. This is precisely what the missing multi-grader ceiling (and the absence of spectroscopic confirmation for most candidates) prevents us from resolving — the gap is the result.

6 Spectroscopic pre-grading

On the gold spectroscopic set the catalog+imaging grader re-grades 20/20 of the 20 Hsu Table-2 Grade-A pairs as plausible lenses and rejects 4/4 of the 4 Foundry-II confirmed non-lenses, using the SIS θ_{E}/separation consistency and the DR10 imaging.

7 The Foundry-I modelability criterion

The graders so far apply the Huang-2020 discovery rubric (imaging) and Hsu-2025 + physical-consistency reasoning (spectroscopy). The one Foundry-paper criterion runnable on a grz cutout is Foundry-I's (Huang 2025a): *does a plausible lens MODEL reproduce the configuration?* We implement it as `quick_lensmodel` — a real GIGA-Lens MAP fit (EPL+shear mass, two Sérsic lens-lights, Sérsic+shapelet source) run per cutout in the GIGA-Lens venv ($\sim$45 s, GPU, cached), keeping JAX out of the SDK process. The discriminator is a `lens_score` combining the recovered θ_{E} and the χ^2 improvement from ray-tracing the source through the mass (large for a true lens, $\sim$0 for a smooth galaxy). The agent can call it as an extra tool (it does: verified end-to-end).

Crucially, evaluating it answers whether a *physical* check beats the vision graders on the hard pool — and it does not. On a balanced validation set the `lens_score` separates real lenses from *ordinary* galaxies well (AUC 0.77) but fails to separate them from the *hard Grade-D human-rejects* (AUC 0.44; the rejects' mean score is slightly *higher* than the lenses'). A held-out-slice run corroborates the pattern: `lens_score` is flat across graded-B/C, Grade-D rejects, and random galaxies. The interpretation is the same as for vision: Grade-D rejects are CNN-flagged objects that look lens-like enough to admit an EPL fit, so on ground-based *grz* a lens model reproduces them about as well as it reproduces true lenses. **Vision, a stronger backbone, multi-agent deliberation, and a physical lens-model fit all hit the same wall on the hard pool** — strong, convergent evidence that the limit is the data + the soft labels, not the grader. The modelability tool remains useful in its proper regime: filtering real lenses from ordinary galaxies (the broad finder pool), not from already-vetted hard rejects.

8 Engineered representations (RepresentationKit)

The graders so far feed the model the raw Lupton-RGB and let it infer the lensing geometry. We instead built a kit of physically-motivated transforms that make the signal explicit and emit scalar lensing-features (`common/representations.py`; the `lens_representations` tool): improved lens-light subtraction, a polar/tangential (r, θ) transform (an arc becomes a horizontal bar), a 180° symmetry residual (a counter-image pair lights up), a blue-excess color-isolation map at the Einstein radius, and a DoG+Hessian "arcness" ridge. Each feature is scored as a standalone discriminator (`eval/run_representations.py`) over a larger 545-system manifest with bootstrap CIs, on three contrasts.

Table 2: RepresentationKit standalone feature AUC (best single / logistic combo), with the agent rich-view arm. Consensus-referenced; no human ceiling.

Contrast	best feature	combo
HARD: graded A/B/C vs Grade-D human-reject	0.57	0.51
EASY: graded A vs ordinary galaxy	0.72	0.70
GOLD: Foundry-II confirmed vs non-lens	0.80(tangential/parity)	—
SILVER: noiseless sims, lensed vs unlensed	0.80(arcness/asym.)	—
Agent (48-slice): lean RGB AUC 0.44 vs +representations 0.40		

The kit closes the question cleanly. The features *demonstrably* extract lensing signal: on noiseless sims arcness/asymmetry reach AUC 0.80, and on spectroscopically *confirmed* labels the tangential-extent and counter-image-parity features reach 0.80. They also give a cheap easy-regime pre-screen (combo 0.70, ∼50 ms vs the 45 s GIGA-Lens fit). But on the *hard consensus pool* they collapse to combo 0.51 — and that does not change with cleaner Tier-2 implementations (photutils isophote, skimage Frangi, SEP geometry: all ≤ 0.57) *nor* when the representations are fed to the agent (0.40 vs lean 0.44, at +43% cost). The *same features* score 0.80 on confirmed labels and 0.51 on the consensus pool — the sharpest evidence in this report that the hard-pool wall is a **label/data limit, not an algorithm limit**: vision, a stronger backbone, deliberation, a GIGA-Lens fit, hand-engineered representations (Tier-1 and Tier-2), and the agent reading those representations *all* land at chance on the soft Grade-D pool, while every method that is tested against clean (confirmed or simulated) labels succeeds.

9 Caveats and what is not done

(1) **No human ceiling** — all labels are single consensus; the multi-grader study is future work. (2) **Grade-D negatives are lens-like** (CNN-flagged), so the binary AUC is lower — and more honest — than against random galaxies; strata are scored separately. (3) The **CNN pre-filter gate** (a cost lever at full-survey scale) is not exercised here because the eval set is by construction the high-p_{meta} ambiguous middle band. (4) Orchestration of the §9.1 specialists is *programmatic* rather than LLM-dispatched `AgentDefinition` subagents, for reliability. (5) Spectroscopic raw-flux streaming is a documented enhancement.

10 Conclusions

LensJudge shows an agentic VI grader can be *built* on the Claude Agent SDK and, more importantly, *tested* on data already in hand. The test returns a nuanced answer. The agent is strong where the task has hard labels and clean evidence — spectroscopic discordant-redshift pairs (20/20 Grade-A plausible, 4/4 non-lenses rejected). On imaging grading of the CNN's hard high-score pool it is near-chance against the single-consensus A/B/C labels, and — the load-bearing result — *no* stronger backbone or multi-agent design changes that. Because those labels are single-consensus over mostly-unconfirmed candidates, we cannot say whether the agent or the consensus is closer to truth: the missing multi-grader ceiling is not a footnote but the thing that decides the question. Two concrete next steps follow directly: (i) collect a $\sim$250-cutout, $\geq$2-grader study to establish the human-vs-human κ and re-anchor every imaging number; (ii) push agentic effort toward the *spectroscopic* channel and toward candidates with confirmation, where the evidence is decisive and the agent already performs. A pre-filter that confidently clears obvious non-lenses while escalating the genuinely ambiguous — which is what LensJudge does — remains useful even at chance agreement on the subjective middle, but its accuracy cannot be claimed until the ceiling exists.

References

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